COMICCRAFT

AI Comic Story Creator

4. System Design & Architecture

Team Information

Team Member	Role
Joseph Thomas	Team Leader
Sugeeve Raj	Team Member
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Architecture Flow

FastAPI receives user input → Gemini generates a structured five-panel outline → Gemini expands the story with narration/captions/dialogue → an image-generation backend creates one illustration per panel → the layout builder associates images with story panels → FPDF2 exports the comic PDF → Jinja2 renders the web pages.

Major Components

- app/main.py – creates the FastAPI application, mounts static files, includes routes, and provides /health.
- app/config.py – manages environment-based settings and storage directories.
- app/schemas.py – defines PromptRequest, PanelOutline, StoryPanel, and ComicLayout.
- gemini_flash.py – creates the five-panel outline.
- gemini_pro.py – expands the outline into a polished comic story.
- image_generator.py – supports demo, Hugging Face, and optional local Diffusers generation.
- layout_builder.py – combines story panels with generated image paths.
- exporters.py – creates the final PDF using FPDF2.
- Jinja2 templates – provide homepage, comic preview, and export-success pages.

Project Structure

The documented project contains app/, app/services/, app/static/, app/templates/, tests/, .env.example, .gitignore, README.md, and requirements.txt.